Evaluating the Impact of Narrative-Based Augmented Reality on Early STEM Learning Outcomes in Sri Lankan Primary Education

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Sri Lankan Grade 4 and Grade 5 students often underperform in mathematics and science, especially in topics that require visual and conceptual understanding. This research project develops four independent narrative based mobile Augmented Reality storybook applications that focus on measurement, life science, physical science, and number concepts. Each application addresses a specific gap in the primary school curriculum by providing bilingual Sinhala and English AR learning content through affordable Android devices. This proposal focuses on Component 4, the AR Number Concepts Storybook, which addresses fractions, decimals, and place value. These topics are difficult for many students because they are commonly taught through abstract symbols with little visual explanation. The proposed application uses the Unity engine and the Vuforia AR framework to display interactive three-dimensional scenes that appear when students scan printed storybook pages. Learners can explore number concepts by grouping, dividing, and counting virtual objects within a simple narrative environment presented in Sinhala and English. Student understanding will be measured before and after instruction and compared with a control group that learns through conventional textbook based methods.

Keywords: Augmented Reality, Number Concepts, Primary Mathematics Education, Narrative-Based Learning

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LIST OF ABBREVIATION

Abbreviation	Definition
AR	Augmented Reality
STEM	Science, Technology, Engineering, and Mathematics
SDK	Software Development Kit
UI	User Interface
UX	User Experience
NIE	National Institute of Education
MOE	Ministry of Education
SDG	Sustainable Development Goal

1. INTRODUCTION

1.1 Background and Context

Mathematics is a foundational subject in primary education, and a strong understanding of number concepts during the early years is essential for long-term academic success. In Sri Lanka, the national mathematics curriculum for Grade 4 and Grade 5 introduces students to topics such as fractions, decimals, place value and basic arithmetic operations. These concepts form the basis for more advanced mathematical reasoning that students will encounter in secondary education and beyond. However, evidence from national assessments, particularly the Grade 5 Scholarship Examination, consistently indicates that a significant proportion of students perform poorly on questions related to fractions and number relationships.

The primary reason for this difficulty is the way number concepts are taught in conventional classroom settings. Teachers typically introduce fractions as symbolic expressions such as 1/2 or 3/4 and expect students to perform operations on these symbols without first establishing a visual or concrete understanding of what the symbols represent. As a result, many students learn to manipulate numerical symbols through memorized procedures rather than developing a meaningful conceptual understanding. For example, students may correctly calculate that 1/4 plus 1/4 equals 2/4 but may not understand that this represents combining two equal parts of a whole. Similarly, the connection between the decimal 0.5 and the fraction 1/2 often remains unclear when these concepts are taught in isolation using textbook diagrams alone.

Place value presents a comparable challenge. Students are expected to understand that a number such as 247 is composed of 2 hundreds, 4 tens, and 7 units. Without physical or visual models that demonstrate how individual units combine to form tens and hundreds, students frequently treat multi-digit numbers as sequences of separate digits rather than as structured quantities. This shallow understanding leads to errors in addition, subtraction, and comparison of numbers.

These difficulties are not unique to Sri Lanka. Research across multiple countries has shown that primary school students worldwide struggle with the transition from concrete arithmetic to abstract number concepts. However, the challenge is particularly pronounced in resource-constrained educational environments where classrooms have limited access to physical manipulatives, interactive teaching aids, and technology-enhanced learning tools. In many Sri Lankan government schools, especially in rural areas, the primary instructional resources are the textbook and the blackboard.

Augmented Reality (AR) has emerged as a promising technology for addressing these challenges in STEM education. AR overlays digital content, such as three-dimensional models, animations, and interactive elements, onto the physical environment when viewed through a smartphone or tablet camera. In educational applications, AR enables students to visualize abstract concepts in a tangible and interactive manner. A meta-analysis by Chang et al. [7] covering ten years of AR research in education found that AR consistently improved learning outcomes in subjects requiring visualization, including mathematics and science. Similarly, Tian et al. [8] reported significant improvements in student cognition and academic performance when AR was integrated into classroom instruction.

Several studies have specifically examined the use of AR in primary mathematics education. Wangid, Rudyanto, and Gunartati [1] demonstrated that AR-assisted storybooks reduced mathematical anxiety and increased student engagement. Rahman and Wangid [2] explored the combination of narrative learning environments with AR visualizations to support mathematics concept understanding. Lintiasri [3] reported improved mathematics learning outcomes among fifth-grade students who used AR-based media compared with those who received traditional instruction. These findings suggest that AR can serve as an effective bridge between abstract mathematical symbols and the concrete understanding that primary school students require.

Despite this growing body of evidence, there is currently no AR learning system specifically designed for the Sri Lankan primary school mathematics curriculum. Existing AR mathematics applications have been developed for other national curricula and are typically available only in English. No bilingual Sinhala and English AR learning materials exist for number concepts at the Grade 4 to Grade 5 level. Furthermore, most existing AR mathematics tools present isolated exercises rather than embedding mathematical concepts within a narrative context that provides motivation and real-world relevance.

This research proposes the development and evaluation of a narrative-based AR learning storybook that focuses on number concepts for Sri Lankan Grade 4 and Grade 5 students. The system generates and presents an interactive storybook in which mathematical concepts arise naturally within everyday situations. When students scan a page of the printed storybook using a mobile device, interactive three-dimensional AR visualizations appear and allow the student to explore fractions, decimals, place value and basic operations through direct manipulation.

The proposed solution aligns with the United Nations Sustainable Development Goal 4 (Quality Education), which aims to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. Specifically, SDG Target 4.1 calls for all children to complete free, equitable, and quality primary education leading to relevant and effective learning outcomes. By providing an accessible, bilingual, and technology-enhanced learning tool that can function on standard mobile devices, this project seeks to contribute to improved mathematics education outcomes in Sri Lankan primary schools.

This research falls within the Interactive Media research cluster, as it involves the design, development, and evaluation of an interactive AR application that combines narrative design, three-dimensional visualization, and user interaction to support educational objectives.

1.2 Literature Review

A growing body of research has examined the use of AR in primary mathematics education, and a number of key studies provide a direct foundation for the proposed system.

Wangid, Rudyanto, and Gunartati [1] investigated the effectiveness of AR-assisted storybooks in reducing mathematical anxiety among elementary school students. Their findings indicated that integrating AR with narrative learning environments can lower student anxiety and increase engagement in mathematics lessons. This study is directly relevant because the proposed system similarly combines narrative storybook delivery with AR visualization.

Rahman and Wangid [2] explored the use of storybook-based AR in mathematics education at the primary school level. Their research demonstrated that narrative and visual interaction can support concept understanding, particularly for students who struggle with abstract symbolic representations. The storybook format adopted in the current proposal builds on this established framework.

Lintiasri [3] studied the effect of AR media on mathematics learning outcomes among fifth-grade students. The research found that AR-based learning media produced significantly improved student performance compared with traditional teaching methods, providing direct empirical support for the expected outcomes of the proposed system.

Saputro, Purnasari, Lumbantobing, and Sadewo [4] reviewed how AR supports different learning styles and enhances student engagement through interactive three-dimensional visualisation of mathematical concepts. Their review highlighted the particular benefit of AR for learners who struggle with two-dimensional textbook representations, which is a central concern of the present research.

Ozdemir [5] examined AR-supported activities in mathematics classrooms and reported improvements in both student motivation and conceptual understanding. The study reinforces the argument that motivation and understanding are closely connected when students interact with visual and manipulable representations.

Koparan [6] developed AR materials for mathematics learning and documented positive responses from both teachers and students regarding engagement and concept visualisation. The design approach in that study, which prioritised subject-specific AR content over generic applications, informs the curriculum-aligned design strategy adopted in the current proposal.

Chang and colleagues [7] conducted a meta-analysis of ten years of AR in education and confirmed that AR consistently improves learning outcomes, particularly in subjects requiring visualisation including mathematics and science. The breadth of this meta-analysis provides strong justification for the AR approach adopted in this research.

Tian and colleagues [8] reported improvements in student cognition and academic performance when AR was integrated into classroom instruction, with particular gains observed in tasks requiring spatial and relational reasoning. Number concepts involving fractions and place value require precisely this type of reasoning.

Yanuarto and colleagues [9] found that AR-based learning environments increase student motivation and interest in mathematics, particularly for learners who have previously experienced difficulty with the subject. Motivation is a significant concern in the Sri Lankan primary mathematics context, where low performance can lead to disengagement.

Ahmad and Junaini [10] conducted a systematic literature review on AR for mathematics learning and identified improved visualisation and conceptual understanding as the most consistently reported benefits. Their review also noted that the majority of existing AR systems for mathematics have been developed for contexts outside South Asia, which directly highlights the gap that the current research addresses.

1.3 Research Gap

The literature review reveals several significant gaps in the current body of research on AR-assisted mathematics education for primary school students. While multiple studies have demonstrated the general effectiveness of AR in improving mathematics learning outcomes, the existing solutions exhibit important limitations that the present research aims to address.

First, no existing AR mathematics learning tool has been specifically designed for the Sri Lankan national curriculum. The studies reviewed were conducted in other educational contexts, including Indonesia, Turkey, and China, and the AR content was developed to align with the respective national curricula of those countries. As a result, the mathematical concepts, examples, and progression sequences used in those tools do not directly correspond to the topics and learning objectives specified in the Sri Lankan Grade 4 and Grade 5 mathematics syllabus.

Second, none of the reviewed studies offer bilingual learning materials in Sinhala and English. Sri Lankan classrooms include students who learn primarily in Sinhala and those who learn in English, and effective educational technology must accommodate both language groups. The absence of bilingual AR materials means that existing tools cannot serve the Sri Lankan student population without significant adaptation.

Third, while several studies have explored AR in mathematics education, very few have combined narrative-based learning with interactive three-dimensional visualization of number concepts. Most existing AR mathematics applications present isolated exercises or single-topic demonstrations. The integration of a continuous story narrative with AR visualizations that cover fractions, decimals, place value, and basic operations within a single coherent system has not been attempted in the existing literature.

Fourth, the majority of existing studies focus on general mathematics topics or geometry, with limited attention to number concepts specifically. Fractions, decimals, and place value are among the most challenging topics for primary school students, yet targeted AR interventions for these specific concepts remain scarce.

Main Research Question

Can a narrative-based mobile AR system improve STEM concept understanding and knowledge retention in Sri Lankan Grade 4–5 students across mathematics and science topics compared to conventional textbook-based instruction?

Sub-Components Addressing the Research Question

- **SRQ1:** Does a narrative-driven AR measurement experience improve conceptual understanding and practical application accuracy for measurement topics (length, mass, capacity, time, and data) among Sri Lankan Grade 4–5 students compared to conventional textbook-based instruction?
- **SRQ2:** Does a narrative AR storybook depicting biological life cycles and ecosystem processes improve scientific process understanding and knowledge retention in Sri Lankan Grade 4–5 students compared to static diagram-based textbook instruction?
- **SRQ3:** Does an interactive AR physical science storybook improve conceptual understanding of light, shadow, and forces and reduce persistent misconceptions in Sri Lankan Grade 4–5 students compared to teacher-led demonstration with static diagrams?
- **SRQ4**: Does a narrative AR storybook embedding fraction, decimal, and number operation concepts in a real-world context improve fraction conceptual understanding and number operation accuracy in Sri Lankan Grade 4–5 students compared to conventional symbolic instruction? (This proposal focuses on SRQ4.)

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate a set of four narrative-based mobile Augmented Reality storybook applications that improve STEM concept understanding and knowledge retention among Sri Lankan Grade 4–5 students across mathematics and science topics, compared to conventional textbook-based classroom instruction.

2.2 Specific Objectives

- To develop a narrative-driven AR measurement storybook application that makes length, mass, capacity, time, and data concepts visually concrete for Grade 4–5 mathematics students through interactive 3D measurement instrument simulations. (Component 1)
- To develop a narrative-driven AR life science storybook application that animates biological processes — plant life cycles, food chains, and ecosystems — using locally grounded Sri Lankan species to improve scientific understanding in Grade 4–5 Parisaraya students. (Component 2)
- To develop a narrative-driven AR physical science storybook application that makes light ray propagation, shadow formation, and force dynamics spatially visible and interactively manipulable for Grade 4–5 Parisaraya students, targeting the misconceptions that arise from static, diagram-only instruction. (Component 3)
- To develop a narrative-driven AR number concepts storybook application that embeds fractions, decimals, place value, and number operations within a real-world bakery context to improve number concept understanding and operational accuracy in Grade 4–5 mathematics students. (Component 4 — this proposal)

3. METHODOLOGY

This research adopts a Design Science Research approach, which is appropriate for studies that involve the design, development, and evaluation of a novel artefact intended to solve a well-defined practical problem [6,7]. The research proceeds through three primary phases: analysis and design, development and testing, and experimental evaluation.

3.1 Research Approach

The analysis phase begins with a structured review of the Sri Lankan Grade 4 and Grade 5 mathematics curriculum to identify the specific number concepts that will be addressed. This review is supplemented by a diagnostic assessment administered to a sample of target students to confirm the areas of greatest conceptual difficulty [3]. The findings directly inform the content scope and interaction design of the AR storybook system.

The development phase applies an iterative prototyping model. Initial low-fidelity prototypes of the storybook pages and AR scenes are reviewed by subject matter experts, including primary school mathematics teachers and educational technology specialists [6]. Feedback from each review cycle is incorporated into subsequent iterations until a stable prototype is achieved. The finalised system is then subjected to technical testing to verify AR trigger accuracy, application stability, and bilingual content correctness.

3.2 Technical Methods and Algorithms

The AR system employs image-based target recognition, in which each printed storybook page functions as a visual marker [4]. When the mobile device camera detects a page within the application, the Vuforia SDK identifies the target image and triggers the corresponding three-dimensional AR scene in real time. The Unity game engine manages scene rendering, student interaction, and feedback logic.

Interactive behaviors within each AR scene are implemented through touch-based event handling. For fractions, students shade or select equal parts of a divided object. For place value, three-dimensional unit blocks combine dynamically to form tens and hundreds. For arithmetic operations, objects group and separate in response to student input [4]. Immediate visual and audio feedback is provided after each interaction to confirm correct or incorrect responses [3].

3.3 Tools and Platforms

Unity 2022 LTS is selected as the development environment because of its strong support for mobile AR development, its integration with the Vuforia AR SDK, and its compatibility with Android [4,6]. Vuforia is selected for image target recognition because of its proven performance in printed-material-triggered AR applications. Adobe Illustrator is used for storybook page design, and Blender is used for three-dimensional model creation. All AR labels and instructions are produced in both Sinhala and English using Unicode-compliant Sinhala fonts.

3.4 Data Collection and Ethical Considerations

The experimental evaluation employs a pre-test and post-test quasi-experimental design [7,8]. A treatment group uses the AR storybook system for a defined instructional period, while a control group receives conventional textbook-based instruction covering the same content. Both groups complete identical pre-test and post-test instruments measuring conceptual understanding and mathematical accuracy. A usability questionnaire is administered to both students and their teachers following the intervention [9].

All data collection activities will be conducted with the written informed consent of school administrators and the parents or guardians of participating students. Student identities will be anonymized throughout data analysis and reporting. The study will be reviewed and approved by the relevant institutional ethics committee prior to data collection.

3.5 Validation Strategy and Performance Metrics

The primary performance metric is the pre-test to post-test gain score in mathematical accuracy, measured using a curriculum-aligned assessment instrument [3,7]. Secondary metrics include task completion rate within the AR system, response accuracy per AR interaction type, and usability ratings collected through a validated instrument [9]. AR system performance is evaluated on trigger detection accuracy, scene loading time, and application crash frequency. A statistically significant improvement in gain scores in the treatment group relative to the control group will be taken as evidence of the effectiveness of the AR storybook system [8].

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system is structured around four principal components that interact to deliver AR-enhanced number concept learning through a printed storybook interface: the printed storybook interface, the mobile AR application, the AR content engine, and the bilingual content layer.

4.1 System Overview

The printed storybook serves as the physical entry point for student interaction. Each page contains a narrative passage that introduces a number concept in a contextually meaningful situation, along with a printed image target that the mobile application uses to identify and trigger the corresponding AR scene. When a student opens the mobile application and points the device camera at a storybook page, the Vuforia image recognition engine identifies the target and passes scene identifier data to the Unity AR content engine, which renders the appropriate three-dimensional AR visualization overlaid on the physical page.

The AR content engine manages five categories of interactive scenes corresponding to the five number concept domains: fractions, decimals, place value, and basic arithmetic operations. Each scene is stored as a self-contained Unity scene asset. The interaction manager within each scene handles touch input from the student, evaluates the correctness of the response, and triggers the appropriate feedback response. Feedback is delivered through visual animation and audio confirmation in both Sinhala and English.

The bilingual content layer operates across all scenes and storybook text. Sinhala and English labels are managed through a localization asset that can be toggled by the teacher or student at the application level. All audio instructions are pre-recorded in both languages.

4.2 Component Interactions and Data Flow

The data flow within the system begins when the device camera captures a video frame. The Vuforia SDK processes the frame and compares detected features against the stored image target database. Upon a successful match, Vuforia returns the target identifier and estimated pose to the Unity AR manager. The Unity AR manager uses the identifier to load the corresponding scene and positions the AR content relative to the detected target pose. Student touch inputs are captured by the Unity event system, processed by the scene interaction manager, and evaluated against the expected answer. The result triggers the appropriate audio and visual feedback response. No student interaction data is transmitted to external servers; all processing occurs on the local device.

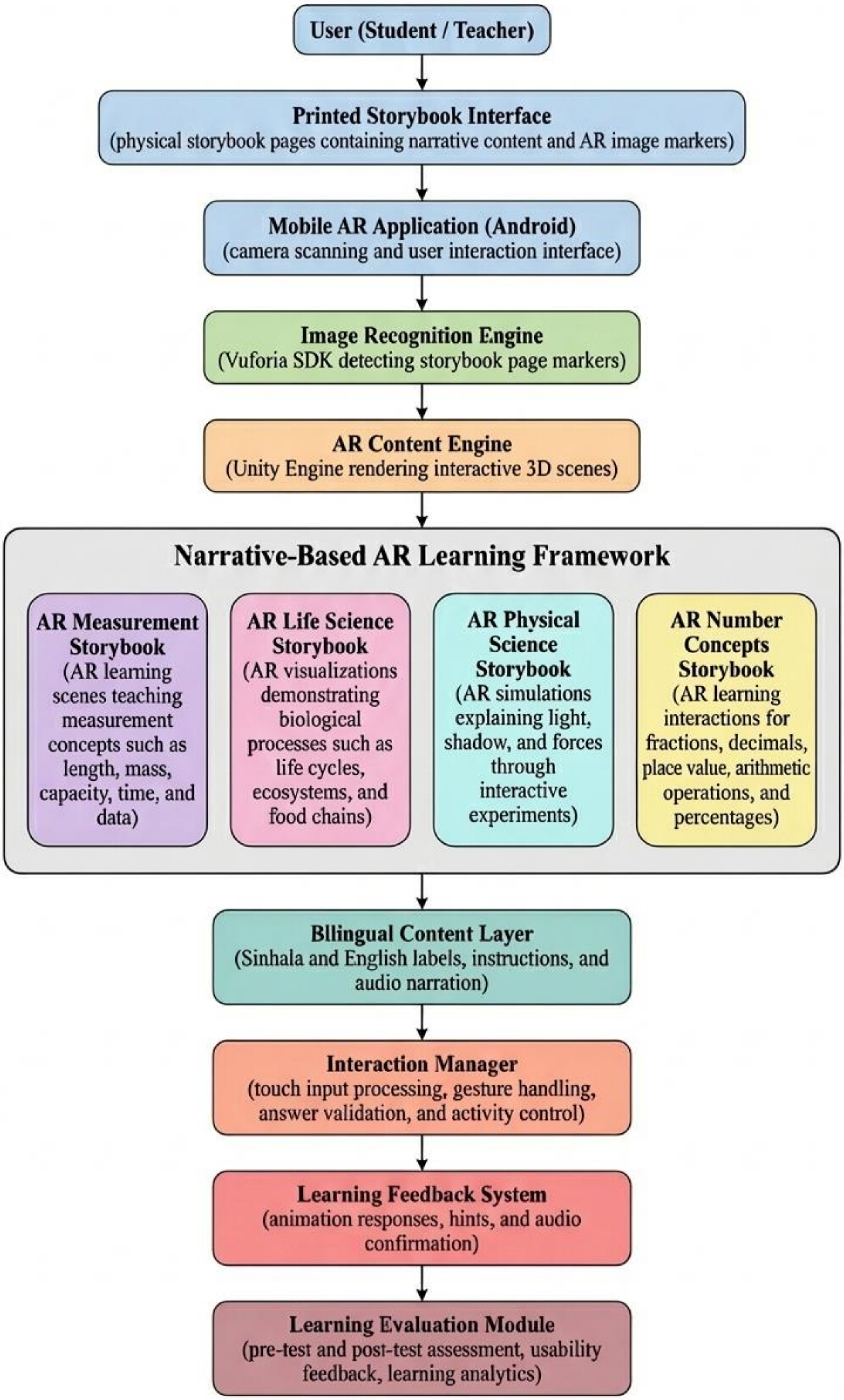

Figure 1: System Overview Diagram

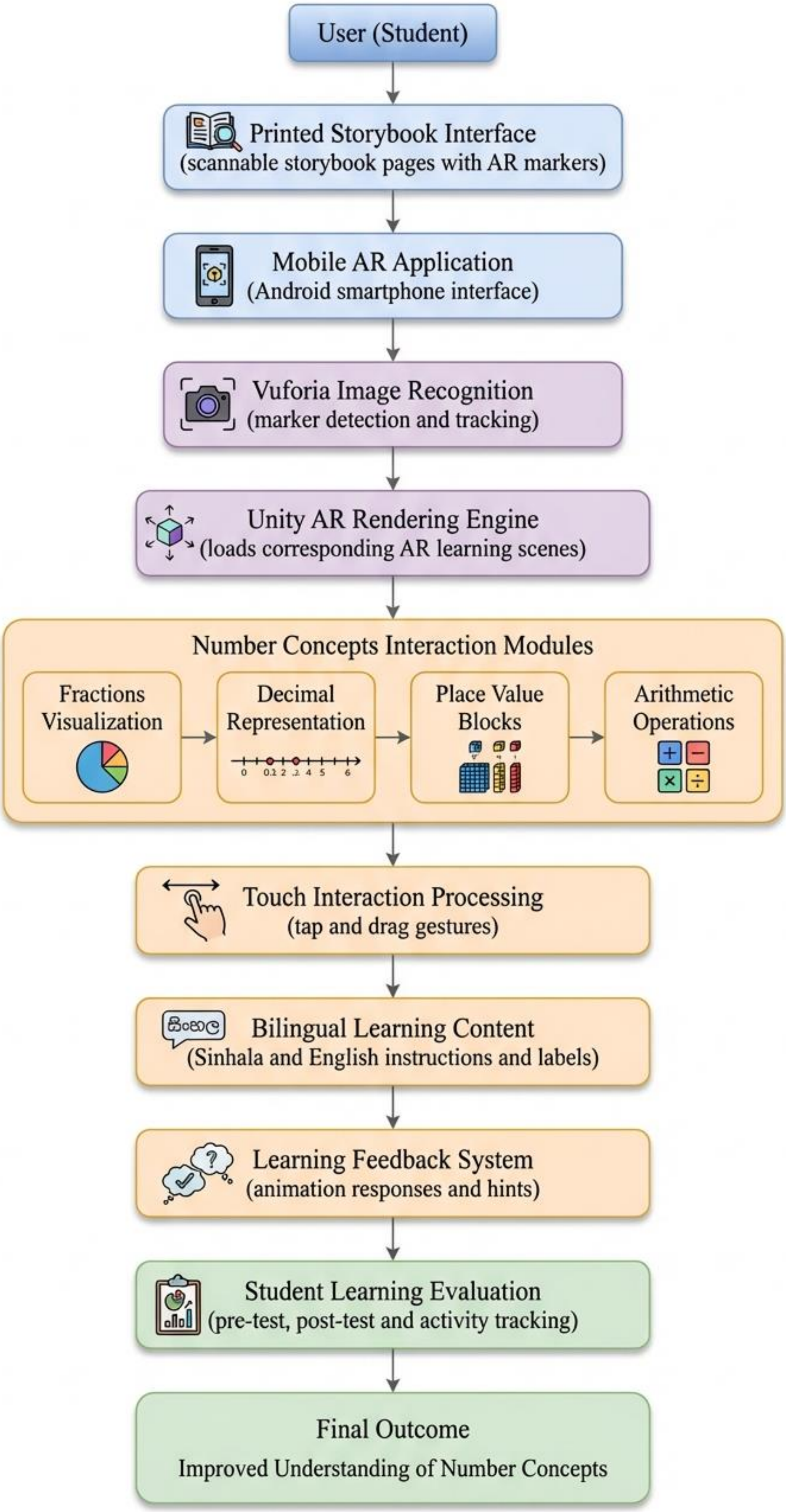

Figure 2: Component Overview

4.2 Scalability and Security

The system architecture supports the future addition of new storybook pages and AR scenes by adding new image target entries to the Vuforia target database and new scene assets to the Unity project without modifying the core application logic. All content assets are bundled within the application package, which eliminates the need for an internet connection during student use. This design is appropriate for deployment in Sri Lankan primary schools, where internet connectivity may be limited or unreliable.

5. USER REQUIREMENTS

5.1 Scalability and Security

- Primary Students (Grade 4 and Grade 5): The direct end users of the AR storybook system who interact with the printed storybook and the mobile application during learning activities.
- Primary School Mathematics Teachers: Secondary users who deploy the system in classroom settings, observe student interaction, and provide feedback on usability and curriculum alignment.
- School Administrators: Institutional stakeholders who approve the use of the system within their schools and facilitate access to student participants for the evaluation study.
- Parents and Guardians: Stakeholders who must provide informed consent for student participation and whose support is necessary for home-based use of the application.

5.2 Functional Requirements

- **AR Scene Triggering:** The system shall detect printed storybook page image targets and trigger the corresponding AR scene promptly upon recognition under normal lighting conditions
- **Fraction Visualization:** The system shall render interactive three-dimensional models that divide objects into equal parts and allow students to select or shade the correct parts to represent a given fraction
- **Decimal Visualization:** The system shall display decimal numbers alongside their fraction equivalents in AR so that students can observe and understand their numerical relationship
- **Place Value Visualization:** The system shall present animated three-dimensional unit block models that combine dynamically to form tens and hundreds, helping students visualize positional value
- **Arithmetic Operations:** The system shall animate the grouping and separation of objects to demonstrate addition, subtraction, multiplication, and division in a visually clear and interactive manner
- **Immediate Feedback:** The system shall provide visual and audio confirmation of correct and incorrect student responses immediately after each interaction
- **Bilingual Interface:** All AR labels, instructions, and audio prompts shall be available in both Sinhala and English, with a language toggle accessible from the main application menu.
- **Touch Interaction:** The system shall support intuitive touch-based student interaction, including tap and drag gestures, within all AR scenes
- **Offline Operation:** The system shall function fully without an internet connection after the initial installation on the student device, ensuring reliable use in schools with limited connectivity.

5.3 Non-Functional Requirements

- **Performance:** The system shall load AR scenes responsively on budget Android devices commonly available in Sri Lankan schools, without noticeable delay that would disrupt the learning experience
- **Usability:** The interface shall be simple and intuitive enough for students grade 4 to 5 to navigate without adult assistance after a brief orientation session
- **Reliability:** The application shall operate stably throughout a standard classroom session without crashing or freezing under normal usage conditions.
- **Security:** No personally identifiable student data shall be collected or transmitted by the application. All interaction processing shall occur locally on the device.
- **Accessibility:** All audio content shall include on-screen text equivalents in both Sinhala and English to support students with hearing difficulties.
- **AR Detection Robustness:** The image target recognition system shall perform reliably under typical classroom lighting conditions, including both natural and artificial indoor lighting.
- **Portability:** The application shall be compatible with commonly available Android smartphones and tablets found in Sri Lankan primary school environments.
- **Maintainability:** The system shall be structured to allow new AR scenes and storybook content to be added in future iterations without requiring significant changes to the core application logic.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Primary market: Sri Lankan government and private primary schools seeking supplementary STEM learning tools for Grades 4–5. Secondary market: Tuition classes and private educational institutes that prepare students for the Grade 5 Scholarship Examination, where physical science is a tested subject. Tertiary market: Parents of Grade 4–5 students looking for interactive home learning tools for science preparation. Extended market: EdTech distributors serving South and Southeast Asian primary education sectors with bilingual AR content requirements.

6.2 Value Proposition

The application provides the only AR physical science learning tool designed specifically for the Sri Lankan Grade 4–5 Parisaraya curriculum, available in both Sinhala and English, deployable on affordable Android smartphones without internet dependency, and validated through empirical student outcome data.

6.3 Revenue Model

- School and Institute Licensing: Annual license fee for schools, tuition centres, and educational institutes covering unlimited student use within one institution.
- Storybook Sales: Printed storybooks sold as physical consumables per student, creating a recurring revenue stream independent of the software license.
- Content Expansion Packs: Additional AR scene packs for other Parisaraya topics sold as add-on licenses to existing institutional customers.

6.4 Cost Estimation

The initial development cost covers software licenses, hardware for testing, storybook design and printing, and developer labour across the research phase. After the research phase, ongoing costs are expected to be low — primarily covering cloud hosting, periodic content updates, and storybook reprinting for new student cohorts. The printed storybook is the main recurring physical cost, as each student requires their own copy to use as an AR marker. Digital distribution of the application itself through the Google Play Store has a one-time registration fee, after which updates can be pushed at no additional distribution cost.

6.5 Pricing Strategy

- Institutional License (Schools and Tuition Institutes): A flat annual license fee per institution, covering unlimited student use of the application within that institution. This makes the cost predictable for school administrators and removes any per-student barrier to access.
- Storybook: Sold separately as a physical printed book per student. Keeping the storybook affordable is important since it is a required physical component of the system. pricing will be set to be comparable to existing school workbooks available in the Sri Lankan market.
- Freemium Entry Option: A free version of the application covering one storybook chapter may be offered to schools and institutes to demonstrate the product before committing to a full license, lowering the adoption barrier.
- Bundle Pricing: Schools that adopt all four components of the broader AR STEM system (Measurement, Life Science, Physical Science, and Number Concepts) would be offered a discounted bundle rate compared to purchasing each component license individually.

6.6 Competitive Advantage

No direct competitor exists for AR physical science instruction in the Sri Lankan primary market. The combination of curriculum alignment, bilingual content, storybook narrative design, and offline capability creates a defensible market position. The empirical validation component provides evidence-based differentiation from unvalidated edtech tools.

6.7 Intellectual Property Considerations

The storybook narrative, character designs, and AR scene assets are original works created for this project. Copyright will be registered with the National Intellectual Property Office of Sri Lanka. The Unity + Vuforia technical stack is used under standard commercial licenses. A patent application for the narrative-AR page-trigger interaction design may be considered upon completion of the research phase.

7. BUDGET AND JUSTIFICATION

Item	Estimated Cost (LKR)	Justification
Storybook Printing (Prototype Copies x 10)	8,000	Physical storybook prototypes are required for AR image target testing and student evaluation activities.
Unity Personal License	0	Unity Personal is available at no cost for educational and non-commercial research projects.
Vuforia SDK License (Developer Tier)	0	The Vuforia Developer licence is free and supports up to 1,000 image targets, sufficient for this prototype.
Blender 3D Modelling Software	0	Blender is an open-source 3D modelling tool adequate for creating all AR scene assets required in this project.
Google Play Developer Account	7,500	A one-time registration fee for publishing the application on the Google Play Store for evaluation distribution.
Audio Recording (Bilingual Voice-Over)	10,000	Professional recording of all Sinhala and English audio instructions and feedback prompts.
Cloud Storage and Version Control	3,000	Used for team asset backup and data collection storage during the evaluation phase.
Human Effort	270,000	Human effort covering approximately 10 months of development and research work for designing, building, testing, and evaluating the AR application.
Miscellaneous	8000	Transport, Internet Contingency
TOTAL	**306,500**	

Table 1: Budget & Justification

8. WORK BREAKDOWN STRUCTURE (WBS)

1. **Literature Review and Gap Analysis:** Review existing research on AR-based mathematics learning, narrative storybook approaches, and AR systems for primary education in developing contexts to identify the gaps this component addresses and ground all design decisions in peer-reviewed evidence.
2. **Storybook Narrative and Page Design:** Design the printed storybook by writing narrative passages that introduce each number concept in a meaningful everyday context, and create the page layouts that will serve as image targets for the AR system, ensuring the visual design is clear and print-ready for a standard school printer.
3. **AR Image Target Preparation and Validation:** Generate a set of candidate AR image target markers from the storybook page designs, evaluate each marker using the Vuforia Target Manager, and shortlist only those that receive a strong recognition rating to ensure reliable detection under typical classroom lighting conditions.
4. **3D Asset and Environment Creation:** Create all 3D models and assets required for the AR scenes using Blender.
5. **Unity Project Setup and Vuforia Integration:** Set up the Unity development environment with the Vuforia SDK and establish a single project structure that can be built and installed on commonly available Android mobile devices without requiring separate configurations for different device types.
6. **AR Scene Development for Number Concepts:** Develop the interactive three-dimensional AR scenes for each number concept domain, including fractions, decimals, place value, arithmetic operation with touch-based interaction and immediate visual and audio feedback in both Sinhala and English.
7. **Bilingual Content Integration:** Integrate all Sinhala and English labels, on-screen instructions, and pre-recorded audio prompts into the AR scenes through a localization asset, and add a language toggle to the main application menu so that teachers and students can switch languages easily.
8. **Performance Optimization:** Optimize the application step by step using asset compression and scene management techniques so that it runs smoothly without lag or freezing on the low-end and mid-range mobile devices commonly found in Sri Lankan school and household environments.
9. **Expert Review and Iterative Refinement:** Present the prototype storybook and AR application to primary school mathematics teachers and educational technology specialists, collect structured feedback on content accuracy, usability, and curriculum alignment, and refine the system across multiple iterations based on that feedback.

10. **Pre-Test and Post-Test Instrument Development:** Design and validate the assessment instruments used to measure student conceptual understanding and mathematical accuracy before and after using the AR storybook system, ensuring alignment with the Sri Lankan Grade 4 and Grade 5 mathematics curriculum.

11. **Experimental Evaluation with Student Participants:** Deploy the AR storybook system with a treatment group of Grade 4 and Grade 5 students over a defined instructional period, administer the pre-test and post-test instruments to both the treatment and control groups, and collect usability questionnaire responses from students and teachers.

12. **Data Analysis and Results Compilation** Analyze the pre-test and post-test gain scores statistically to determine whether the AR storybook system produced a significant improvement in conceptual understanding and mathematical accuracy compared with conventional instruction and compile all findings into structured result tables and charts.

13. **Final Report Writing and Submission** Write the complete research report covering all phases of the project, including literature review, methodology, system design, evaluation findings, and conclusions, and submit it by the deadline.

9. GANTT CHART

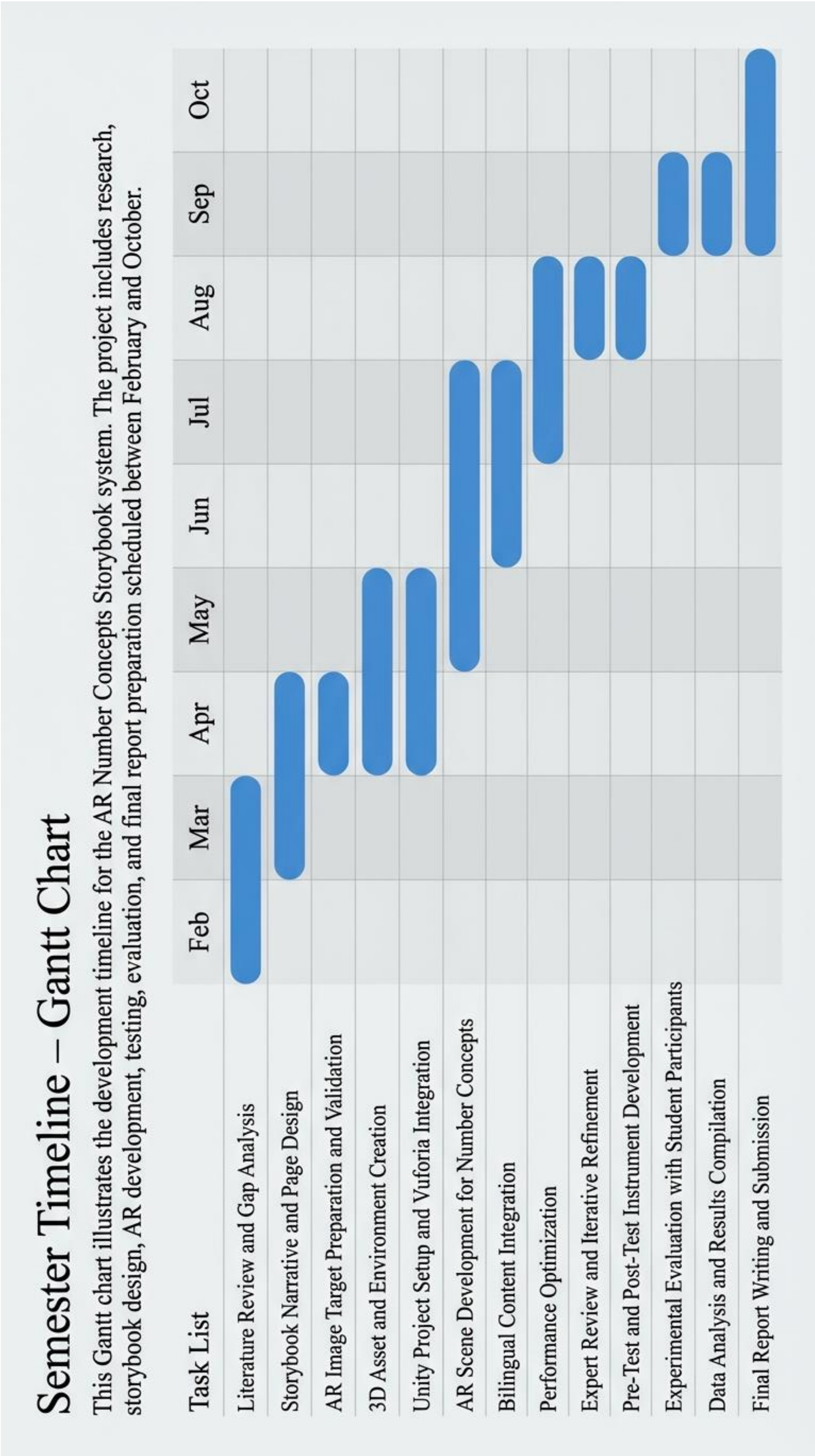

Figure 3: Timeline Gantt Chart

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